Questions

Q1.

A student places a trolley at the top of a runway, as shown in Figure 17.

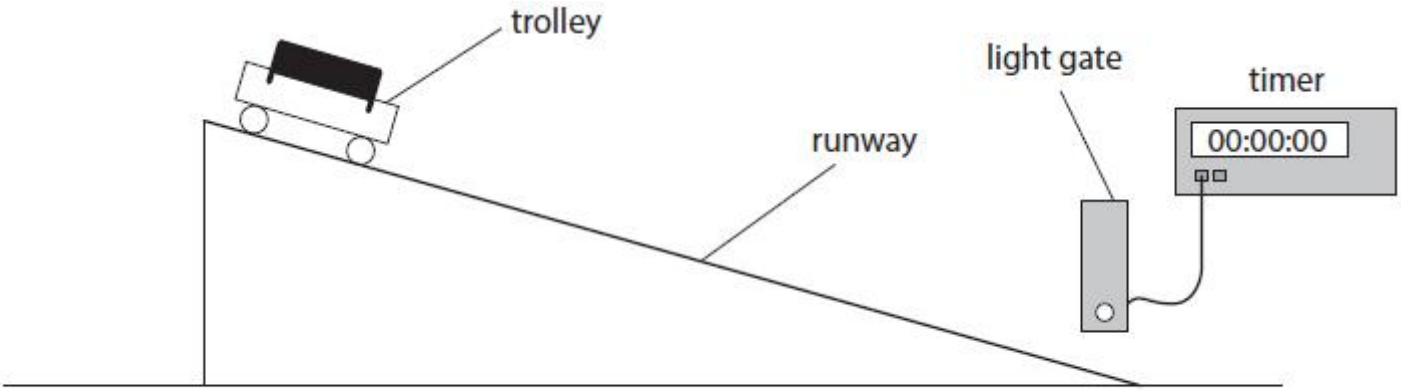

Figure 17

The student had to lift the trolley through a height of 26 cm to put it at the top of the runway.

The gravitational potential energy, Δ*GPE*, of the trolley increases by 4.1 J when lifted to the top of the runway.

The gravitational field strength, *g*, = 10 N/kg.

(i) Show that the mass of the trolley is about 1.6 kg.

Use the equation

$$m = \frac{\Delta GPE}{g \times \Delta h}$$

(2)

(ii) The student releases the trolley and the trolley moves down the runway.

The student assumes that all the extra gravitational potential energy, Δ*GPE*, stored at the top of the runway is transferred to kinetic energy at the end of the runway.
Calculate the expected value of the velocity of the trolley at the end of the runway.
Use the equation

$$v^2 = \frac{2 \times \Delta GPE}{m}$$

(3)

expected velocity = ... m/s

(iii) The light gate, shown in Figure 17, is used to determine the actual velocity of the trolley at the end of the runway.

The actual velocity at the end of the runway is less than the expected velocity as calculated in part (ii). Suggest **one** way to reduce the difference between the actual velocity and the expected velocity.

(1)

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(Total for question = 6 marks)

Q2.

This question is about energy changes.

Figure 12 shows a person pushing a box from the bottom of a slope to the top of the slope.

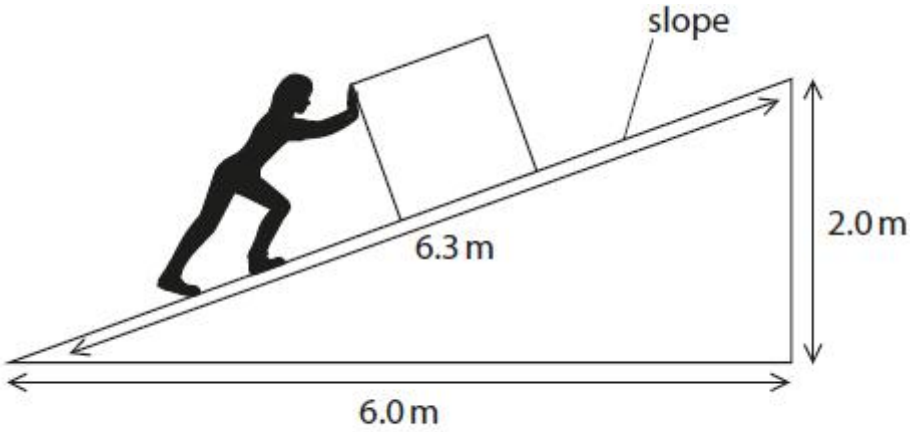

Figure 12

Explain which one of the three distances shown in Figure 12 should be used to calculate the work done against the force of friction between the box and the slope.

(2)

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(Total for question = 2 marks)

Q3.

The system transfers 8400 J of useful kinetic energy to the water passing through the pump in 1 minute.

Figure 13 shows a diagram of the energy transfers.

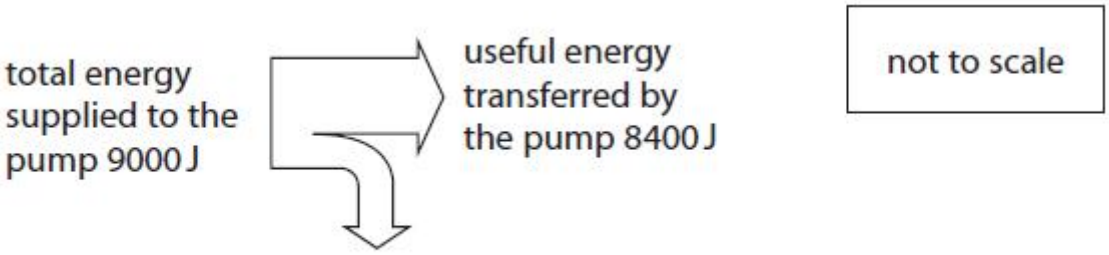

Figure 13

(i) Explain why the useful energy transferred to the water is different from the total energy supplied to the pump.

(2)

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.....

.....

.....

(ii) Calculate the efficiency of the pump.

Use the equation

$$\text{efficiency} = \frac{\text{useful energy transferred by the pump}}{\text{total energy supplied to the pump}}$$

(2)

efficiency = ...

(Total for question = 4 marks)

Q4.

This photograph shows a fan.

The blades of the fan are turned by an electric motor.

In one second, the motor gets 200 J of electrical energy from the mains supply.
Only 180 J of this energy is used to turn the blades of the fan.

The rest of the energy is wasted.

(i) Calculate how much of the 200 J of energy is wasted.

(1)

wasted energy = ... J

(ii) State what happens to the wasted energy.

(1)

.....

.....

(iii) Calculate the efficiency of the motor.

(2)

efficiency = ...

Q5.

A cyclist is riding a bicycle at a steady velocity of 12 m/s.

The cyclist and bicycle have a total mass of 68 kg.

Describe the energy transfers that happen when the cyclist uses the brakes to stop.

(2)

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(Total for question = 2 marks)

Q6.

(i) Complete the equation that relates efficiency, useful energy transferred by a device and total energy supplied to the device.

(1)

efficiency .....

(ii) In one second an engine has a total energy input of 7500 J.

In one second 3200 J is transferred to the surroundings as wasted energy.
Calculate the useful energy transferred by the engine.

(1)

useful energy transferred = ...

(iii) Calculate the efficiency of this engine.

(2)

efficiency of the engine = ...

(Total for question = 4 marks)

Q7.

This question is about energy transfers.

Figure 13 shows the apparatus used for investigating the transfer between gravitational potential energy and kinetic energy.

A metal ball is attached to a thread.

The ball is released from a starting position and swings on the thread.

The ball cuts a light beam at the bottom of its swing.

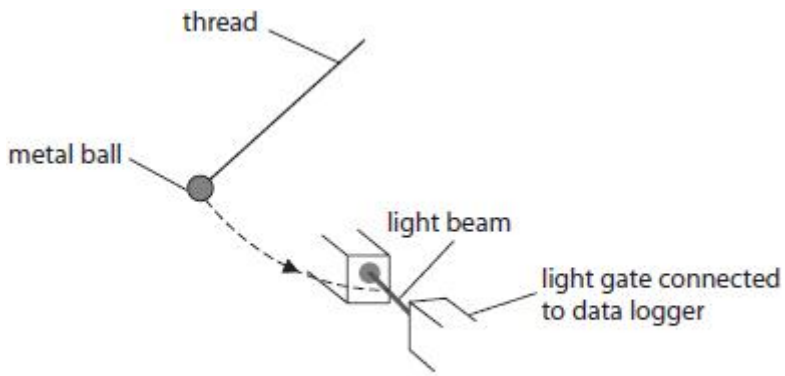

Figure 13

When the ball cuts the light beam, the speed of the ball is recorded by the data logger.

The ball was released 3 times from the same height and the speed measured each time.

The measurements of speed are given in Figure 14.

speed in m/s	1.31	1.27	1.16

Figure 14

Figure 15 shows an end-on view of the ball's swing from its starting position.

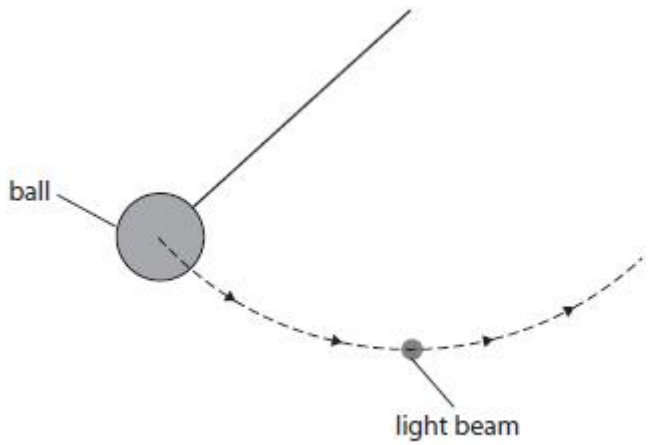

Figure 15

(i) To measure the change in vertical height, Δh, through which the ball moves, a ruler could be used.

Draw a ruler on Figure 15, placed in a position to measure the change in vertical height Δh.

(1)

(ii) Figure 16 shows a set square.

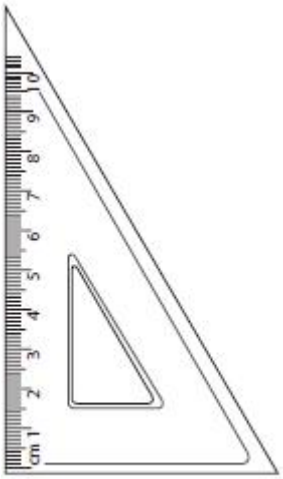

Figure 16

Describe how the measurement of the change in vertical height, Δh, could be improved using the set square.
You may add to Figure 15 or Figure 16 to help your description.

(2)

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(Total for question = 3 marks)

Q8.

*Figure 23 shows a drone.

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Figure 23

The blades on the drone are turned by electric motors.

The electric motors are powered by a battery.

Figure 25 represents the energy transfers involved when the drone rises from the ground.

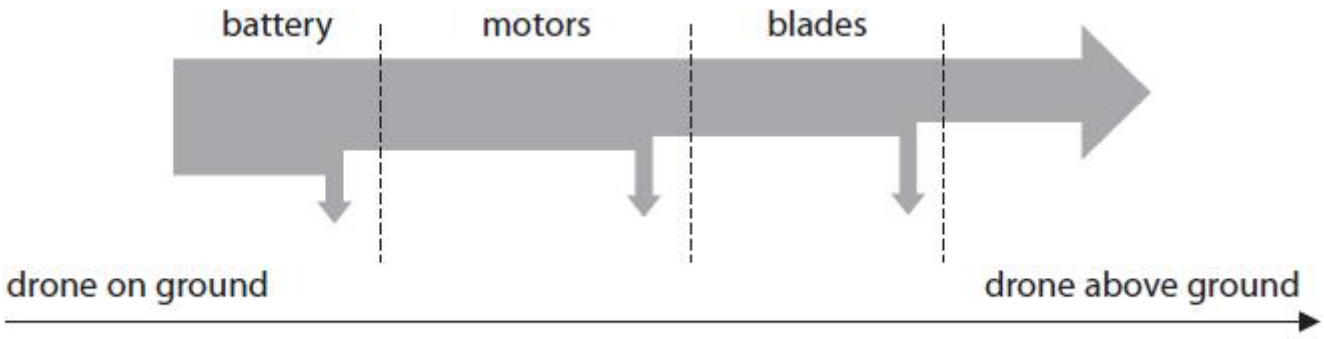

Figure 25

Describe the changes in the way energy is stored when the drone rises from the ground.

Your answer should refer to energy transfers.

(6)

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(Total for question = 6 marks)

Q9.

A student plots a graph showing the height at the start and the maximum height reached after each bounce.

Figure 16 shows the student's graph.

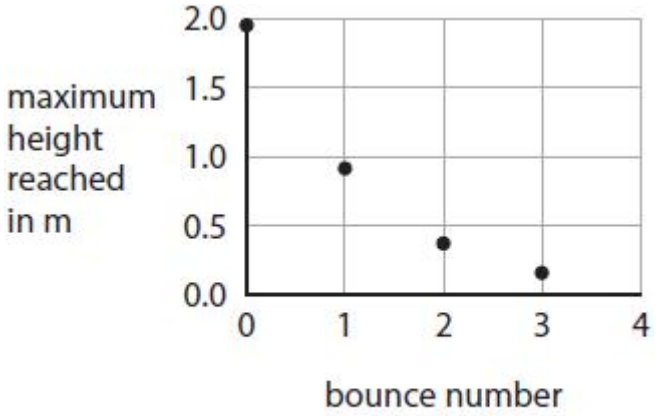

Figure 16

Describe how the maximum height reached changes with the bounce number in Figure 16.

(2)

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(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	substitution with conversion (2) (m =) $\frac{4.1}{10 \times 0.26}$ (=1.58)	accept 9.8(1) instead of 10 for g throughout alternative method substitution and rearrangement with conversion (2) e.g. Δgpe = 10 x 0.26 x 1.6 (=4.1) Allow for 1 mark: (m =) $\frac{4.1}{10 \times 26}$ allow 26 to any power of 10 for this mark OR Δgpe = 10 x 26 x 1.6 allow 26 to any power of 10 for this mark OR height converted to m 0.26 (m) or 26×10^{-2} (m)	**(2)** **AO 2.1**

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) $(v^2=)\ \frac{2 \times 4.1}{1.6}$ evaluation of v^2 (1) 5.1(25) evaluation of v (1) 2.3 (m/s)	ecf from part a(i) throughout accept use of 1.58 accept 5.2 (use of 1.58) accept values that round to 2.3 e.g. 2.264 award full marks for the correct answer without working	(3) AO2.1

Question Number	Answer	Additional guidance	Mark
(iii)	addition of any lubricant / oil / grease	make runway steep(er) use an air track instead streamline trolley	(1) AO 3.3b

Q2.

Question number	Answer	Additional guidance	Mark
	Explanation linking **two** from: choice of distance (1) 6.3 m (calculations of work done need) the distance moved in the direction of the force (1) (friction acts) along the slope / hypotenuse (1)	 accept pushed up the slope	(2) AO3

Q3.

Question number	Answer	Additional guidance	Mark
(i)	An explanation linking energy has been dissipated /wasted / lost (1) as thermal energy (1)	 energy has been transferred mechanically useful energy is less than total energy supplied identifies difference of 600(J) heat / to the surroundings ignore sound accept (some) energy has been transferred to thermal store for 2 marks	(2) AO3

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) (efficiency =) $\frac{8400}{9000}$ evaluation (1) (efficiency =) 0.93	 0.9 93(%) allow values that round to 0.93 or 93(%) award full marks for the correct answer without working	(2) AO2

Q4.

Question Number	Answer	Acceptable answers	Mark
(i)	20 (J)	200 – 180 (even if calculated value from this is incorrect)	**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	(changed to) {thermal energy / heat}	dissipated (lost) to {surroundings / motor / air / atmosphere} sound / noise **reject** if kinetic, light or chemical is mentioned	**(1)**

Question Number	Answer	Acceptable answers	Mark
(iii)	$\frac{180}{200} \times 100$ (1) 90 (%) (1)	award full marks for correct answer with no working $\frac{180}{200}$ 0.9, 9/10 Or [100 – (20/200)] % not needed but if a unit is given then maximum score is 1	**(2)**

Q5.

Question Number:	Answer	Additional guidance	Mark
	a description to include: kinetic energy (store) (of cyclist and /or bicycle) decreases / is transferred into(1) thermal energy (store) (of brakes / surroundings) increases (1)	 KE for kinetic energy allow heat for thermal allow brakes get hotter ignore sound energy accept kinetic (energy) to heat (energy) for 2 marks in this context	**(2)** AO 1 1

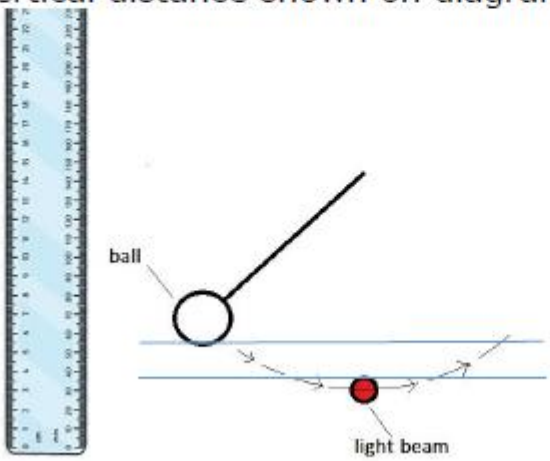
Q6.

Question Number:	Answer	Mark
(i)	efficiency = $\frac{\textbf{useful}\text{ (energy transferred by the device) (x100)}}{\text{total (energy supplied to the device)}}$	**(1)** AO 1 1

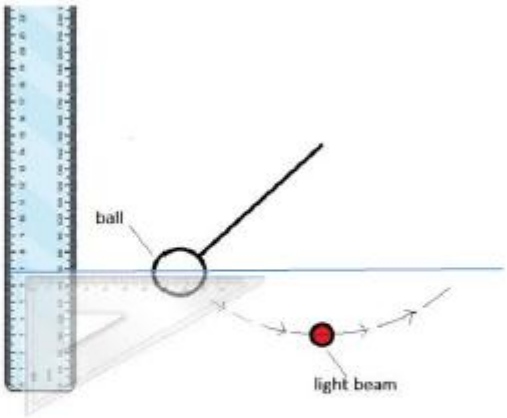
Question Number:	Answer	Additional guidance	Mark
(ii)	determine useful energy (1) 7500 -3200 = 4300		**(1)** AO 2 1

Question Number:	Answer	Additional guidance	Mark
(iii)	substitution (1) efficiency = $\frac{4300}{7500}$ evaluation (1) 0.57	 allow ECF from (i) and/ or (ii) for 1 mark maximum accept 57(.33)(%), 0.6, 60(%) award full marks for the correct answer without working	**(2)** AO 2 1

Q7.

Question number	Answer	Additional guidance	Mark
i	ruler / line / rectangle shown vertically, must include minimum vertical distance shown on diagram (1) 	judge by eye accept any vertical line covering the minimum vertical distance	**(1)** **AO3**

Question number	Answer	Additional guidance	Mark
ii	description to include set square placed against ruler (to measure vertical position) (1) (one edge of set square placed at) right angles / perpendicular / 90° (to ruler) (1) (set square used to) make ruler vertical (1)	accept reasonable alternatives on a diagram or explained in writing accept one edge of the set square shown as vertical in diagram	**(2)** **AO3**

		full marks may be awarded from additions to Figure 15 or 16 e.g.  allow 2 marks for any horizontal line (set square use) on the diagram drawn through / touching a vertical ruler if no other mark scored allow 1 mark for improving accuracy	

Q8.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. • Chemical energy stored in battery • Transferred to KE in motors • Transferred to GPE as it rises • Thermal energy wasted (at each stage) • Energy transferred to surroundings (at each stage)	(6)

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Isolated fact e.g. a description of least one energy store **or** interpretation of diagram without mentioning energy stores or types	Possible candidate responses Chemical energy stored in the battery **or** energy transferred from the battery to the motors and then to the blades. Some energy is lost at each stage.
Level 2	3–4	Additional guidance Description of at least one energy transfer	Possible candidate responses KE (of blades) is transferred to GPE (as the drone rises) **or** (thermal) energy is transferred to the surroundings
Level 3	5–6	Additional guidance Description of two or more energy transfers	Possible candidate responses Chemical energy in the battery is transferred to KE of the blades **AND** Thermal energy is wasted in the motors when they turn.

Q9.

Question Number	Answer	Additional guidance	Mark
	A description to include: as the bounce number increases the height decreases/negative correlation (1) non-linear (1)	 allow not in even steps / not proportional / not a straight line height/it (nearly) halves each time scores 2 marks	**(2)**